

ENSAE PARIS – INSTITUT POLYTECHNIQUE DE PARIS



Research Report — Extension

From Winter Temperatures to Natural-Gas Volatility

*Schwartz–Smith state-space modelling and weather-coupled GARCH
on Henry Hub natural gas*

July 2026

AUTHOR:

Nayel BENABDESADOK

Extension of the report *Stochastic and Machine-Learning Temperature Models for Weather-Derivative Pricing*

SELF-DIRECTED QUANTITATIVE RESEARCH PROJECT

Contents

1	Introduction	1
2	Henry Hub Data and Stylised Facts	2
3	A Structural Detour: the Schwartz–Smith Model	5
3.1	Two Factors for One Price: Specification	5
3.2	From Spot to Futures	6
3.3	State-Space Form and Kalman Filter	6
3.4	Why Estimation Fails with Our Data	7
4	Targeting Volatility: a Weather-Coupled GARCH	9
4.1	The Foundation: a Student-t GARCH(1,1)	9
4.2	Where the Weather Effect Lives: Anomalies, Tails, and Late Winter	10
4.3	The Model Cascade: from Realized Anomaly to Forecast Surprise	12
4.4	Out-of-Sample Validation	13
5	Does a Better Temperature Model Buy Better Gas Volatility?	14
5.1	The design: change only the source of the anticipation	14
5.2	Temperature skill is real, and shows up in sample	14
5.3	Out of sample, the edge evaporates	15
6	Conclusion	16
	References	17
	Appendices	18
A	Moments of the Schwartz–Smith model	19
A.1	Discretization and vector form	19
A.2	Mean and variance by recursion	19
A.3	Continuous-time limit	19
A.4	Log-normality and futures prices	20
A.5	Exact discretization for the filter	20
B	Derivation of the Kalman filter	20
B.1	Prediction	21
B.2	Innovation	21
B.3	Correction and Kalman gain	21
B.4	Likelihood via the prediction error decomposition	22
C	Non-identifiability with a short futures curve	22
C.1	Degeneracy of the model as $\gamma \rightarrow 0$	22
C.2	Orders of magnitude: the condition on maturities	23

1 Introduction

This document extends the main report, *Stochastic and Machine-Learning Temperature Models for Weather-Derivative Pricing*, which built temperature models (Ornstein–Uhlenbeck and machine learning) and pitted them against each other on the pricing of HDD weather derivatives. The present part explores the natural next step: if cold winters inflate heating degree days, they also inflate natural gas demand. Temperature should therefore leave a trace in the price of gas, and it is this trace that we seek to model here on Henry Hub, the US spot market benchmark.

The guiding idea grew out of the main report itself. Degree days emerged there as the variable that summarizes the winter from a desk's point of view, and the pricing contest showed that temperature information, at the seasonal horizon, essentially collapses to climatology. The next question then follows naturally: does this same meteorological information, injected no longer into the payoff of an HDD contract but into the dynamics of the gas underlying, add anything? A desk that hedges weather contracts with gas positions, or the other way around, needs to know whether the two legs react to the same cold shock, and with what intensity.

We attack the problem from two successive angles. The first targets the *level* of the price: a structural two-factor model, that of Schwartz and Smith, estimated by Kalman filter on the futures curve, would have provided a complete risk-neutral framework into which temperature could then be plugged. This route fails, and the failure deserves careful documentation: it is not the equations that break down, but the identifiability of the model when the futures curve is limited to one year of maturities. The second angle targets *volatility*: daily spot returns then suffice, and a GARCH augmented with meteorological variables (degree-day anomalies, forecast surprises) makes it possible to test directly whether cold moves the variance of gas. This route succeeds, and concludes with the exact mirror of the pricing contest of the main report: we plug in the temperature forecasts from the OU and machine-learning models to measure whether a better temperature model makes a better volatility model.

The document follows this path. Section 2 presents the gas data and their stylized facts. Section 3 develops the structural detour: the Schwartz–Smith specification, its state-space form, the Kalman filter, and the identifiability diagnosis, with the complete derivations deferred to the appendices. Section 4 builds the weather-coupled GARCH and tests its out-of-sample predictive contribution. Section 5 finally plugs the temperature models of the main report into this volatility engine, and section 6 concludes.

2 Henry Hub Data and Stylised Facts

The underlying for this entire part is natural gas at the *Henry Hub*, the Louisiana delivery point that serves as the benchmark for US contracts. Two datasets feed the analysis. The daily spot price (FRED, series DHHNGSP, in \$/MMBtu) covers January 1997 to April 2026, a total of 7350 daily returns. The NYMEX futures curve (Databento) provides twelve rolling monthly contracts $C_1, \dots, C_{12}$ from June 2010 to April 2024, whose residual maturities are reconstructed according to the NYMEX expiration convention, three business days before the first day of the delivery month: the maximum available maturity is thus $\tau_{\max} \approx 1$ year, a figure that will play a decisive role in section 3.4.

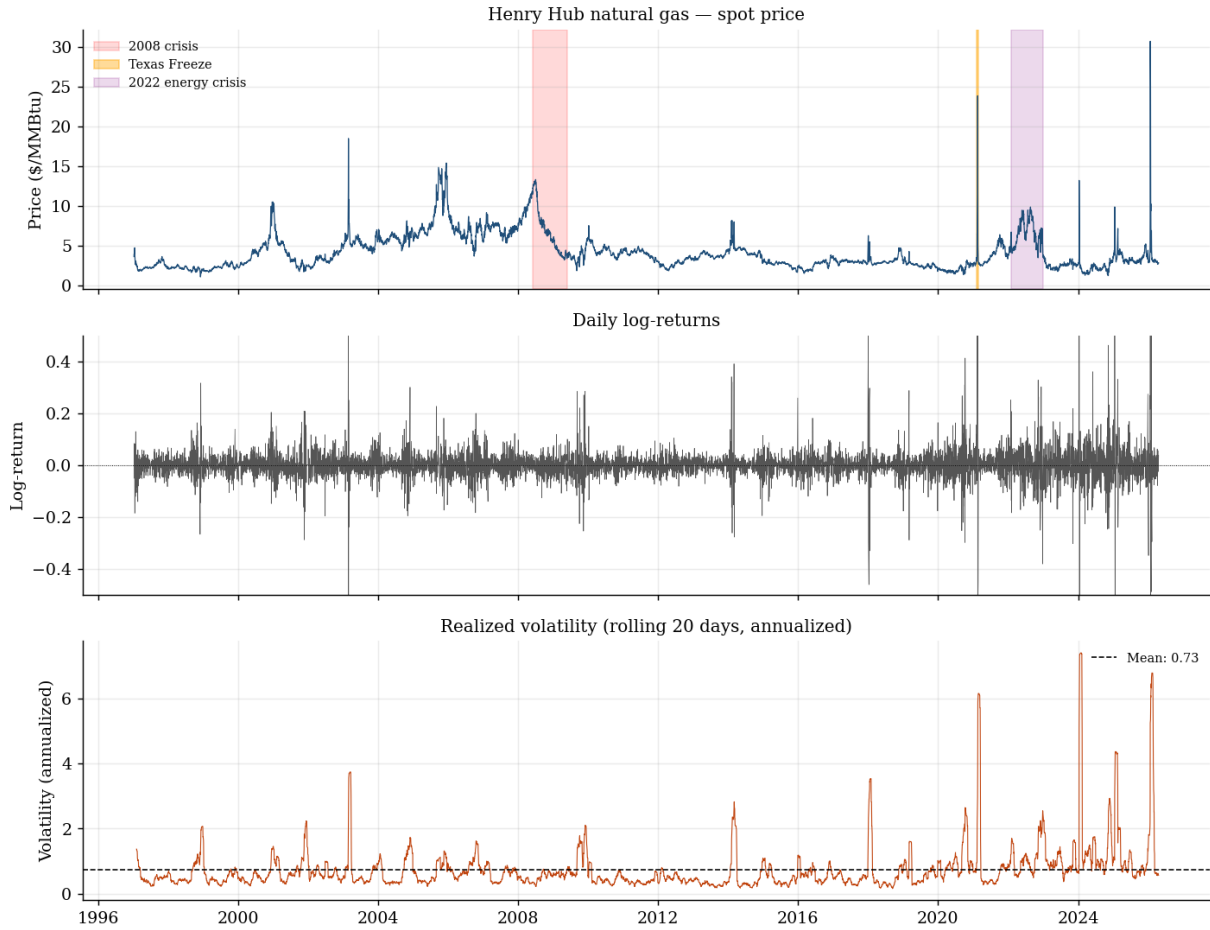


Figure 1: Henry Hub, 1997–2026: spot price (\$/MMBtu) with the major market episodes, daily log returns, and 20-day rolling realized volatility (annualized).

Figure 1 condenses the stylised facts. The price oscillates around a long-run equilibrium level, between 3 and 5 \$/MMBtu for most of the period, from which it departs through violent, asymmetric spikes, systematically upward, and systematically traceable to a weather or geopolitical event: a cold winter on low storage in 2003, hurricanes Katrina and Rita in 2005, the Texas freeze of February 2021, the LNG rush to Europe in 2022. These deviations dissipate within a few weeks. Volatility, moreover, alternates between distinct regimes, turbulent in 2005–2008 and 2021–2023, remarkably calm in 2012–2020: volatility clustering is plain to see, and the ACF of absolute returns (figure 2) confirms it, significant and persistent over more than forty lags, the signature of an autocorrelated conditional variance.

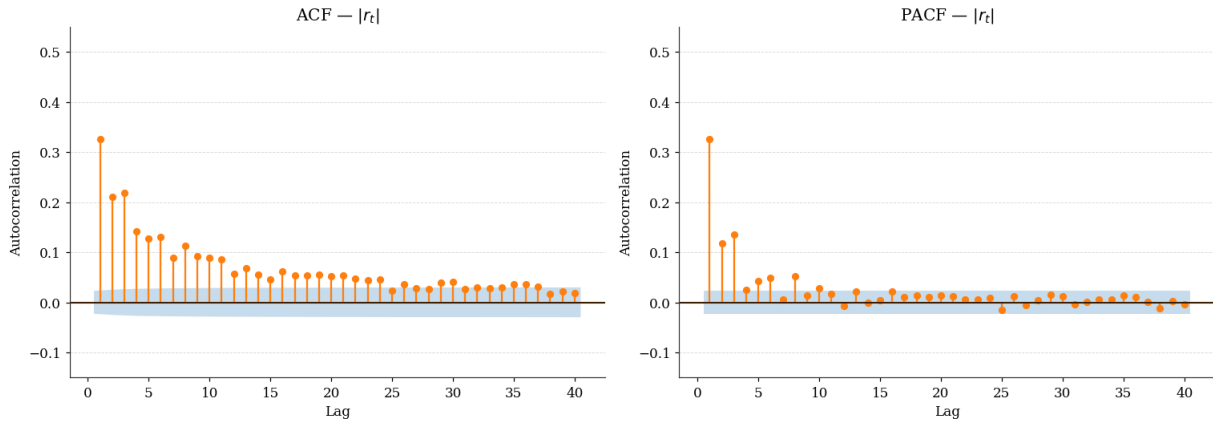


Figure 2: ACF and PACF of the absolute returns $|r_t|$ of the Henry Hub spot, 40 lags.

The distribution of returns is far from Gaussian (figure 3): excess kurtosis of 114.9, positive skewness of 1.10, a Jarque–Bera statistic above 4×10^6 , and a fitted Student’s t distribution whose parameter $\nu \approx 2.2$ borders on the limit of variance existence. The tails are fat and asymmetric, a point that is far from anecdotal for a desk whose payoffs are decided precisely in the tails. Finally, the mean of the returns is indistinguishable from zero (t -test, $p = 0.956$): we will model $r_t = \varepsilon_t$ without a mean term.

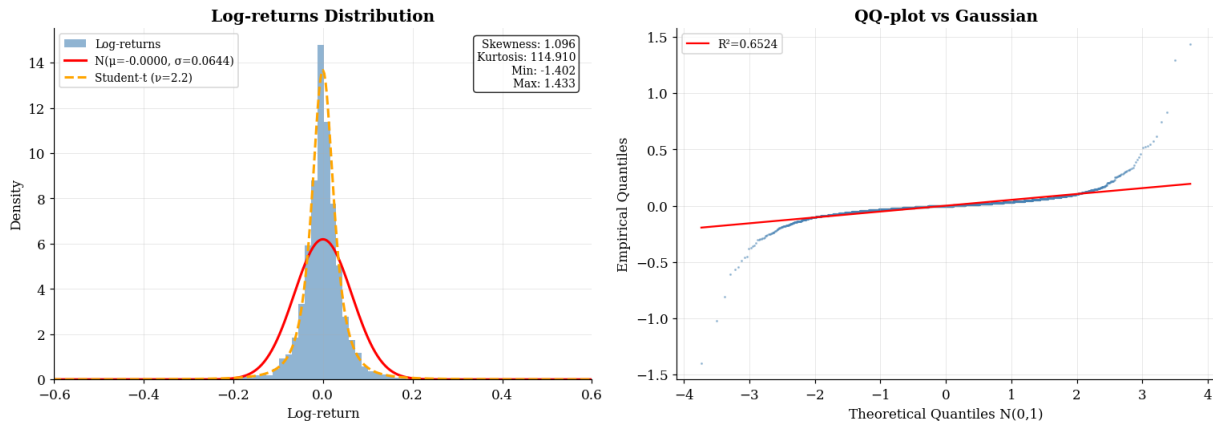


Figure 3: Distribution of Henry Hub spot log returns: histogram, fitted normal and Student’s t densities, and QQ-plot against the Gaussian.

These observations already sketch the structure of the problem: a price level that mixes two time scales (a slow equilibrium and fast transitory shocks) and a variance with a life of its own, persistent, seasonal, and fat-tailed. A Hodrick–Prescott decomposition of the log price, purely illustrative, visualizes the first intuition (figure 4): the smooth trend plays the role of the long-term factor, the cycle that of the transitory deviations. This is exactly the decomposition that the Schwartz–Smith model formalizes, and with which we begin.

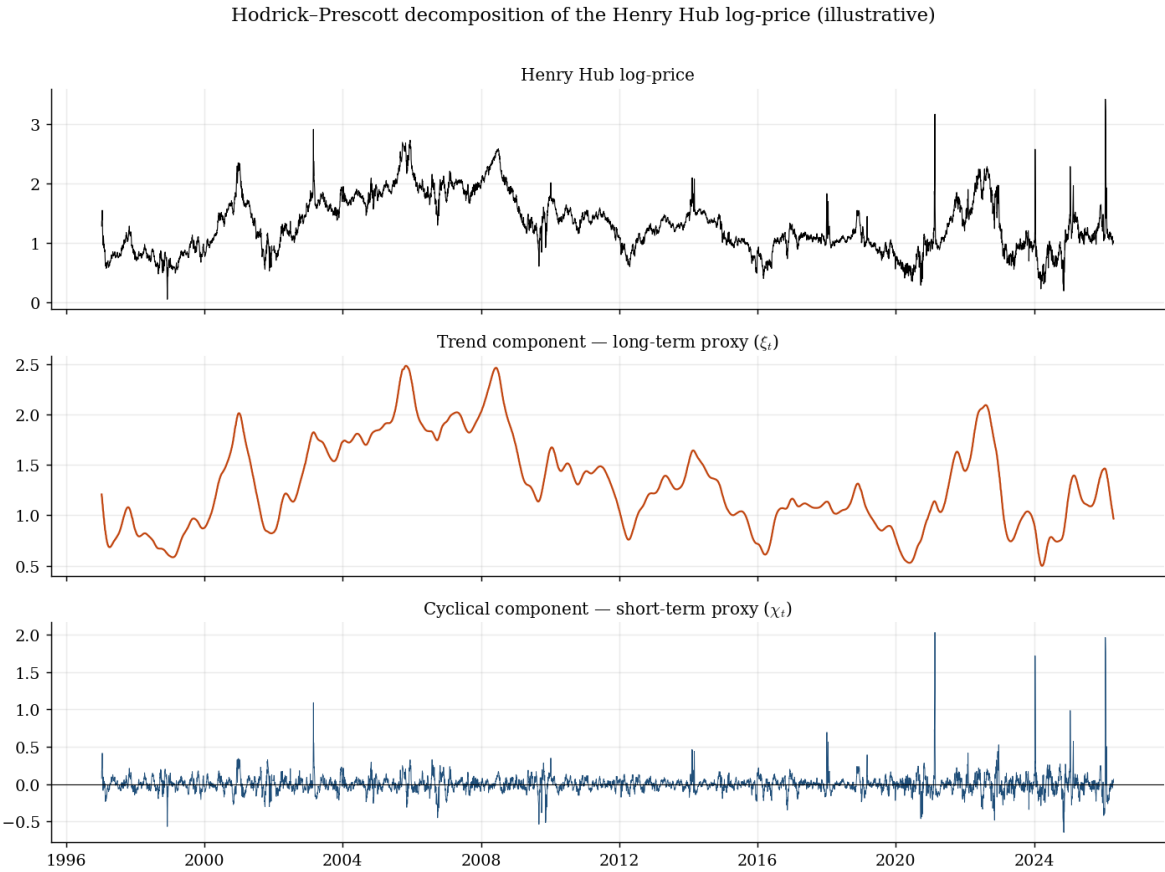


Figure 4: Hodrick–Prescott decomposition of the Henry Hub log price (illustrative): slow trend and transitory cycle.

3 A Structural Detour: the Schwartz–Smith Model

Before tackling volatility, the first attempt was structural: model the *level* of the gas price with a latent-factor model, estimate it on the futures curve, and then plug temperature into it. This route ended in an instructive failure, which this section documents in detail: the model is sound, the implementation is validated, but the available futures data do not allow its parameters to be identified. Understanding precisely why sheds light on the choice, made afterwards, to target volatility rather than the level.

3.1 Two Factors for One Price: Specification

The spot price of gas mixes two dynamics of different natures. In the short run, transitory shocks (a cold snap, supply tightness, a geopolitical event) push the price away from its equilibrium level, then fade over a few weeks or months. In the long run, the equilibrium level itself drifts slowly, driven by structural changes in the market (production capacity, long-term demand). Schwartz and Smith (2000) formalize this intuition by decomposing the logarithm of the spot price S_t into two latent factors:

$$\log S_t = \chi_t + \xi_t, \quad (1)$$

where χ_t carries the short-term dynamics and ξ_t the long-term ones. The short-term factor is a zero-mean Ornstein–Uhlenbeck process, the same mathematical object as our temperature residual:

$$d\chi_t = -\kappa \chi_t dt + \sigma_\chi dZ_t^\chi, \quad (2)$$

with κ the speed of reversion to zero. For the long-term factor, we adopt a specification slightly more general than that of the original paper: a slow mean reversion,

$$d\xi_t = (\mu_\xi - \gamma \xi_t) dt + \sigma_\xi dZ_t^\xi, \quad (3)$$

where γ is small. The canonical model of Schwartz and Smith (2000), in which ξ_t is an arithmetic Brownian motion with drift μ_ξ , is obtained as the limiting case $\gamma \rightarrow 0$. The two Brownian motions are correlated, $dZ_t^\chi dZ_t^\xi = \rho dt$. The economic interpretation is immediate: κ is large (half-life of a few weeks to a few months), γ is close to zero (half-life of several years), and it is precisely this contrast in time scales that, as we will see, makes identification demanding.

Since the state vector $X_t = (\chi_t, \xi_t)^\top$ is Gaussian, its first two moments admit closed forms. Discretizing and then passing to the continuous limit (detailed calculation in appendix A), we obtain

$$\mathbb{E}[X_t] = \begin{pmatrix} e^{-\kappa t} \chi_0 \\ e^{-\gamma t} \xi_0 + \frac{\mu_\xi}{\gamma} (1 - e^{-\gamma t}) \end{pmatrix}, \quad \text{Var}(X_t) = \begin{pmatrix} \frac{\sigma_\chi^2}{2\kappa} (1 - e^{-2\kappa t}) & \frac{\rho \sigma_\chi \sigma_\xi}{\kappa + \gamma} (1 - e^{-(\kappa + \gamma)t}) \\ \frac{\rho \sigma_\chi \sigma_\xi}{\kappa + \gamma} (1 - e^{-(\kappa + \gamma)t}) & \frac{\sigma_\xi^2}{2\gamma} (1 - e^{-2\gamma t}) \end{pmatrix}. \quad (4)$$

Since the logarithm of the spot is the sum of the two factors, it is itself Gaussian:

$$\mathbb{E}[\log S_t] = \chi_0 e^{-\kappa t} + \xi_0 e^{-\gamma t} + \frac{\mu_\xi}{\gamma} (1 - e^{-\gamma t}), \quad \text{Var}(\log S_t) = \frac{1 - e^{-2\kappa t}}{2\kappa} \sigma_\chi^2 + \frac{1 - e^{-2\gamma t}}{2\gamma} \sigma_\xi^2 + 2 \frac{1 - e^{-(\kappa + \gamma)t}}{\kappa + \gamma} \sigma_\chi \sigma_\xi \rho, \quad (5)$$

and S_t is log-normal, with $\mathbb{E}[S_t] = \exp(\mathbb{E}[\log S_t] + \frac{1}{2} \text{Var}(\log S_t))$.

3.2 From Spot to Futures

The factors χ_t and ξ_t are not observed: what the market quotes are futures contracts with maturities $T_1, \dots, T_M$. Under the risk-neutral measure $\mathbb{Q}$ (obtained by subtracting risk premia λ_χ and λ_ξ from the drifts of the two factors), the price of a future with maturity T is $F(t, T) = \mathbb{E}_t^\mathbb{Q}[S_T]$, and log-normality yields a closed form for its logarithm:

$$\log F(t, T) = e^{-\kappa(T-t)} \chi_t + e^{-\gamma(T-t)} \xi_t + A(T-t) \quad (6)$$

where the deterministic function $A(\tau)$ collects risk premia and variance corrections:

$$A(\tau) = -\frac{\lambda_\chi}{\kappa}(1 - e^{-\kappa\tau}) + \frac{\mu_\xi - \lambda_\xi}{\gamma}(1 - e^{-\gamma\tau}) + \frac{1}{2} \left(\frac{1 - e^{-2\kappa\tau}}{2\kappa} \sigma_\chi^2 + \frac{1 - e^{-2\gamma\tau}}{2\gamma} \sigma_\xi^2 + 2 \frac{1 - e^{-(\kappa+\gamma)\tau}}{\kappa + \gamma} \sigma_\chi \sigma_\xi \rho \right). \quad (7)$$

The structure of equation (6) is telling: the short-term factor weighs on short maturities with a weight $e^{-\kappa\tau}$ that dies out quickly, while the long-term factor weighs on the whole curve with a weight $e^{-\gamma\tau}$ that barely decays. Each section of the futures curve therefore carries information about a different mixture of the two factors, and it is this deformation along maturities that should allow them to be separated.

3.3 State-Space Form and Kalman Filter

The model fits naturally into a linear Gaussian state-space form. The exact discretization of equations (2) and (3) (AR(1) scheme, not Euler, consistent with the temperature pillar) gives the state equation

$$X_t = c + G X_{t-1} + \omega_t, \quad c = \begin{pmatrix} 0 \\ \frac{\mu_\xi}{\gamma}(1 - e^{-\gamma\Delta t}) \end{pmatrix}, \quad G = \begin{pmatrix} e^{-\kappa\Delta t} & 0 \\ 0 & e^{-\gamma\Delta t} \end{pmatrix}, \quad (8)$$

where $\omega_t \sim \mathcal{N}(0, W)$ is the state noise, whose covariance is given by the exact integrals of equation (4) taken over a step Δt . The observations are the log-prices of the M contracts quoted at date t , $y_t = (\log F(t, T_1), \dots, \log F(t, T_M))^\top$, linked to the state by the measurement equation, read directly off equation (6):

$$y_t = d_t + F_t^\top X_t + v_t, \quad d_t = (A(\tau_1), \dots, A(\tau_M))^\top, \quad F_t = \begin{pmatrix} e^{-\kappa\tau_1} & \dots & e^{-\kappa\tau_M} \\ e^{-\gamma\tau_1} & \dots & e^{-\gamma\tau_M} \end{pmatrix}, \quad (9)$$

with $\tau_i = T_i - t$ and $v_t \sim \mathcal{N}(0, V)$ a diagonal measurement noise (quotation errors, independent across maturities). The full parameter vector is $\Theta = (\kappa, \gamma, \mu_\xi, \sigma_\chi, \sigma_\xi, \rho, \lambda_\chi, \lambda_\xi, s)$, where s parameterizes V .

Since the state is hidden, estimation relies on the *Kalman filter*: a recursive algorithm that simultaneously solves two problems for a linear Gaussian system with a latent state. At each date, a *prediction* step propagates the estimated state through the dynamics equation (8), then a *correction* step confronts this prediction with the new observation y_t and updates the estimate in proportion to the prediction error, via the *Kalman gain*, which weighs the confidence placed in prediction versus observation according to their respective uncertainties. In the linear Gaussian setting, the filter is optimal (the minimum-variance estimator) and, as a valuable by-product, each prediction error e_t and its covariance $L_{t|t-1}$ provide the contribution of date t to the log-likelihood:

$$\ell(\Theta; y) = -\frac{1}{2} \sum_{t=1}^{M_T} \left[M \log(2\pi) + \log \det(L_{t|t-1}) + e_t^\top L_{t|t-1}^{-1} e_t \right], \quad (10)$$

which is maximized numerically over Θ . The complete derivation of the filter (prediction, innovation, gain, update) is deferred to appendix B.

3.4 Why Estimation Fails with Our Data

The model is estimated by maximum likelihood (L-BFGS-B) on the weekly training data 2010–2018 (447 observations of the twelve-maturity curve), with the risk premia set to zero and the search bounds economically motivated by the plausible half-lives of the two factors, from two weeks to six months for χ_t and from two to twenty years for ξ_t . The symptom is immediate and reproducible: four different initializations all converge to the same log-likelihood (NLL = −30.739), but with parameters pinned against the bounds. The speed κ lands on its lower economic bound ($\hat{\kappa} = 1.403$, a six-month half-life) and, above all, $\hat{\gamma} = 0.0017$ sticks to the lower numerical bound, an implied half-life for the long-term factor of roughly 400 years, which is economically absurd. The optimizer finds no interior minimum: the likelihood is flat in the direction of γ .

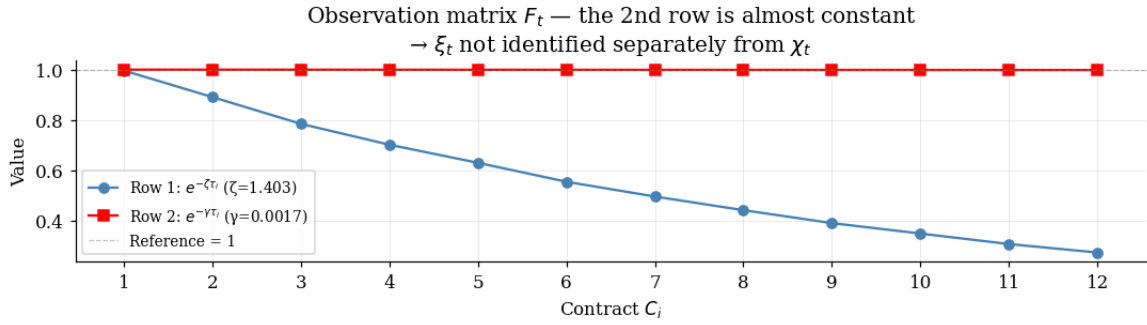


Figure 5: The two rows of the observation matrix F_t at the estimated parameters: $e^{-\kappa\tau_i}$ decays markedly across the twelve contracts, while $e^{-\gamma\tau_i}$ stays flat at 1.

The diagnosis can be read directly off the observation matrix (figure 5). At the estimated parameters, the row of the short-term factor decays visibly along the curve, but that of the long-term factor equals $e^{-\gamma\tau_i} \in [0.99841, 1.00000]$: a total variation of 0.0016 across the twelve maturities. The factor ξ_t loads on all observations identically, so the filter cannot distinguish it from a constant, and its filtered estimate converges to the mean of the futures curve rather than to a structural factor.

The underlying theoretical argument is visible to the naked eye in equation (9). As $\gamma \rightarrow 0$, the second row of F_t tends to a constant vector of ones and the component of $A(\tau)$ carried by the drift becomes linear in τ . Each observation then writes

$$y_{t,i} \approx \left(\mu_\xi + \frac{1}{2} \sigma_\xi^2 \right) \tau_i + g(\tau_i) + e^{-\kappa\tau_i} \chi_t + \xi_t + v_{t,i},$$

where g depends neither on μ_ξ nor on ξ_t . For any $\delta \neq 0$, the pair $(\mu_\xi + \delta, \xi_t - \delta\tau_i)$ reproduces exactly the same observation: the level of the long-term factor and its drift trade off against each other along a direction of the likelihood without anything changing in the distribution of the data. The formal proof of this non-identifiability, in the sense $p(y; \Theta_1) = p(y; \Theta_2)$ with $\Theta_1 \neq \Theta_2$, is given in appendix C: the maximum likelihood estimator is then no longer consistent. This is precisely what $\hat{\gamma}$ pinned against its bound is telling us: our variant with two mean-reverting factors collapses on its own to the degenerate case $\gamma = 0$ of the canonical model.

In practice γ is not exactly zero, and everything hinges on orders of magnitude. To separate the factors, the weight $e^{-\gamma\tau}$ would need to vary substantially over the observed maturity range, that is, $\gamma\tau_{\max} \gg 1$, or $\tau_{\max} \gg 1/\gamma$. With a long-term factor of a plausible half-life of several years ($\gamma \approx 0.1$), this requires maturities on the order of a decade. Yet with $\tau_{\max} = 1$ year, $e^{-\gamma\tau_{\max}} = 0.905$: over our entire strip, the weight of the long-term factor remains confined to $[0.905, 1]$, a variation of less than ten percent, drowned in measurement noise. The short-term factor poses no such problem: with $\kappa \approx 1.3$, $e^{-\kappa\tau_{\max}} = 0.272$, the decay is clearly visible over one year. The identification condition is therefore satisfied for κ but not for γ . This diagnosis is

consistent with the protocol of Schwartz and Smith (2000) themselves, who calibrate their model on futures extending up to nine years.



Figure 6: Filtered states of the Schwartz–Smith model on Henry Hub: short-term factor $\hat{\chi}_t$, long-term factor $\hat{\xi}_t$, and reconstruction of the log-price (red area: out of sample).

The out-of-sample validation drives the point home. The parameters estimated on 2010–2018, held fixed, deteriorate markedly over 2019–2024: the log-likelihood drops from -31.06 to -13.13 and the log-price reconstruction error doubles (RMSE from 0.071 to 0.156). Figure 6 illustrates the trap: the reconstruction $\hat{\chi}_t + \hat{\xi}_t$ visually tracks the log-price very well, including out of sample, but this reconstruction quality validates nothing, since any sufficiently flexible additive model reconstructs whatever series it is given. The objective was the *economic* identification of the two factors, and that is what fails. The details of the optimizer’s convergence, where κ and γ can be seen collapsing onto their bounds, are deferred to figure C.1 (appendix C).

The conclusion of this detour is twofold. First, this is a problem of *data*, not of method: with a long curve (beyond five to ten years of maturities, available via Bloomberg), the same code would identify the model. Second, the failure is rich in lessons for what follows: since the price level resists identification with our data, we shift the target to an object that daily spot data suffice to estimate, its *volatility*.

4 Targeting Volatility: a Weather-Coupled GARCH

The structural detour showed that the price level resists our data. Volatility, by contrast, requires nothing more than daily spot returns, which we have over thirty years. This section builds the volatility model in successive layers: a baseline GARCH first, then exogenous weather variables whose every addition is motivated by a precise empirical diagnostic, up to the final model validated out of sample.

4.1 The Foundation: a Student-t GARCH(1,1)

The stylized facts of section 2 call for the GARCH family: volatility clustering and the persistent ACF of absolute returns are the signature of an autocorrelated conditional variance, exactly what the recursion of Bollerslev (1986), generalizing the ARCH of Engle (1982), is designed to capture. Since the mean of the returns is statistically zero, we set

$$r_t = \varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad z_t \stackrel{iid}{\sim} t_v \text{ standardized}, \quad (11)$$

the Student-t distribution of the innovations being imposed by the fat tails of figure 3. The model is estimated by maximum likelihood on the 1997–2016 training period (5020 observations), with the 2017–2026 period (2330 observations) left untouched for out-of-sample evaluation, a split that deliberately places the Texas freeze and the 2022 energy crisis in the test set.

The estimation yields $\hat{\alpha} = 0.146$ and $\hat{\beta} = 0.825$, both highly significant, implying a persistence of $\hat{\alpha} + \hat{\beta} = 0.971$ and a half-life of variance shocks of about 24 days, with an annualized long-run volatility on the order of 70 %. The choice of the Student-t is settled formally by a likelihood ratio test against the Gaussian ($LR = 458.7$ against a threshold of 3.84), and the estimated degrees of freedom, $\hat{\nu} = 5.6$, confirm markedly fat tails. The diagnostics validate the specification on the variance side: no remaining ARCH effect in the standardized residuals (Ljung–Box on z_t^2 not rejected at all tested lags, Engle’s LM $p = 0.94$), and the conditional volatility tracks realized volatility closely (figure 7).

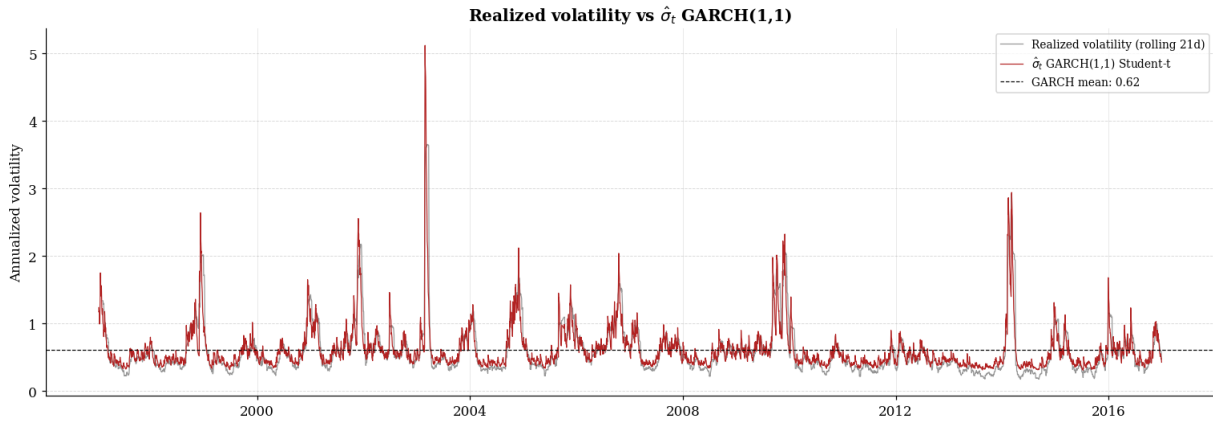


Figure 7: Rolling realized volatility (21 days) and conditional volatility of the Student-t GARCH(1,1), annualized, over the training period.

One limitation remains, and it motivates everything that follows: the three parameters (ω, α, β) know nothing about the calendar. The GARCH absorbs the clustering, but the winter seasonality of volatility, so visible in figure 1, is left unexplained by the endogenous dynamics alone. The natural next step is to inject weather into the variance equation.

4.2 Where the Weather Effect Lives: Anomalies, Tails, and Late Winter

Before adding a variable to the GARCH, one must first construct it correctly. The market fully anticipates the average seasonality: a cold January teaches nobody anything. Only the *surprise* can carry information, which leads to the positive HDD anomaly, the day's excess degree days relative to the 2000–2019 calendar normal, $\widehat{\text{HDD}}_{c,t}^+ = \max(\text{HDD}_{c,t} - \overline{\text{HDD}}_c(m_t, d_t), 0)$. Aggregation across cities uses state-level residential gas consumption (EIA) rather than population, because it is gas heating demand that links cold weather to Henry Hub: the composite weights Chicago at 38.8 %, New York at 38.1 %, Minneapolis at 12.0 %, and Boston at 11.1 %.

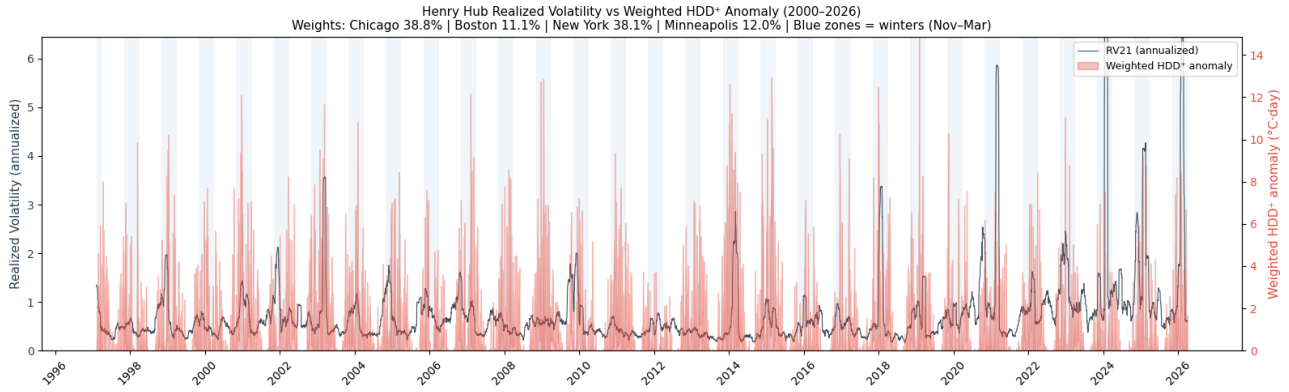


Figure 8: Henry Hub realized volatility (annualized RV21) and weighted composite HDD⁺ anomaly, 1997–2026, with blue bands over the winters (November–March).

Figure 8 sets out the founding puzzle: volatility spikes are almost exclusively wintertime events, but the relationship is not mechanical. Some very cold winters produce no spike at all (2011, 2013), while others explode (2014, 2021). The effect of cold on volatility is therefore conditional, and two analyses locate it precisely. The first is a quantile regression of weekly realized volatility on the composite anomaly, over the winters 2000–2026 (figure 9): the coefficient γ_τ is indistinguishable from zero up to the 80th quantile, then grows sharply beyond it. The weather effect does not shift the center of the volatility distribution; it inflates its tails, precisely the region that governs pricing.

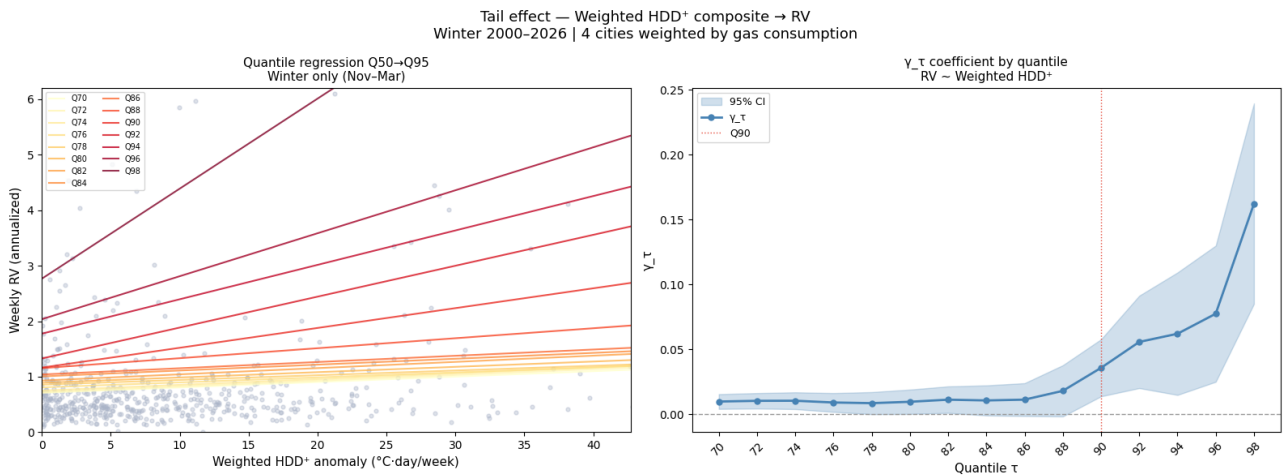


Figure 9: Quantile regression of weekly realized volatility on the composite HDD⁺ anomaly (winters 2000–2026, 579 observations): fitted lines by quantile and coefficient γ_τ with 95 % confidence intervals.

The second analysis slides this regression along the winter in six-week windows (figure 10): γ_{90} is zero in November–December, turns significantly positive from January onward, and peaks in February around 0.10. The same degree day of anomaly is not worth the same volatility depending on the point in the winter, and

the economic explanation can be read off the storage data (figure 11): in early winter, inventories are full and cushion the demand shock, whereas in late winter, storage below its normal level can no longer absorb a cold spell, and the price becomes the only shock absorber. The largest volatility spikes in the sample (2014, 2018, 2022) all correspond to late winters on depleted inventories.

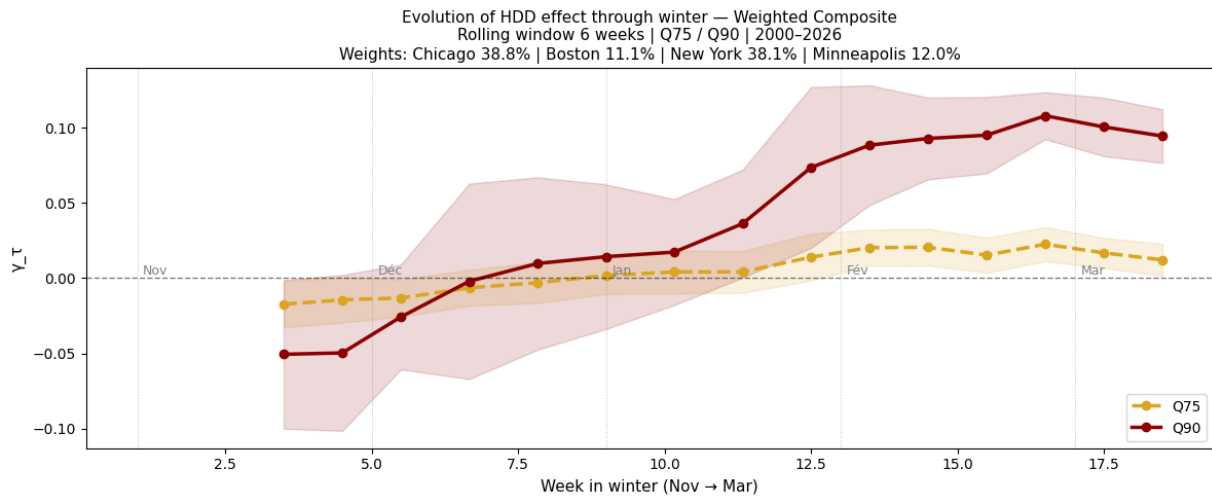


Figure 10: Coefficient γ_τ (75th and 90th quantiles) from the quantile regression over rolling six-week windows along the winter, weighted composite, 2000–2026.

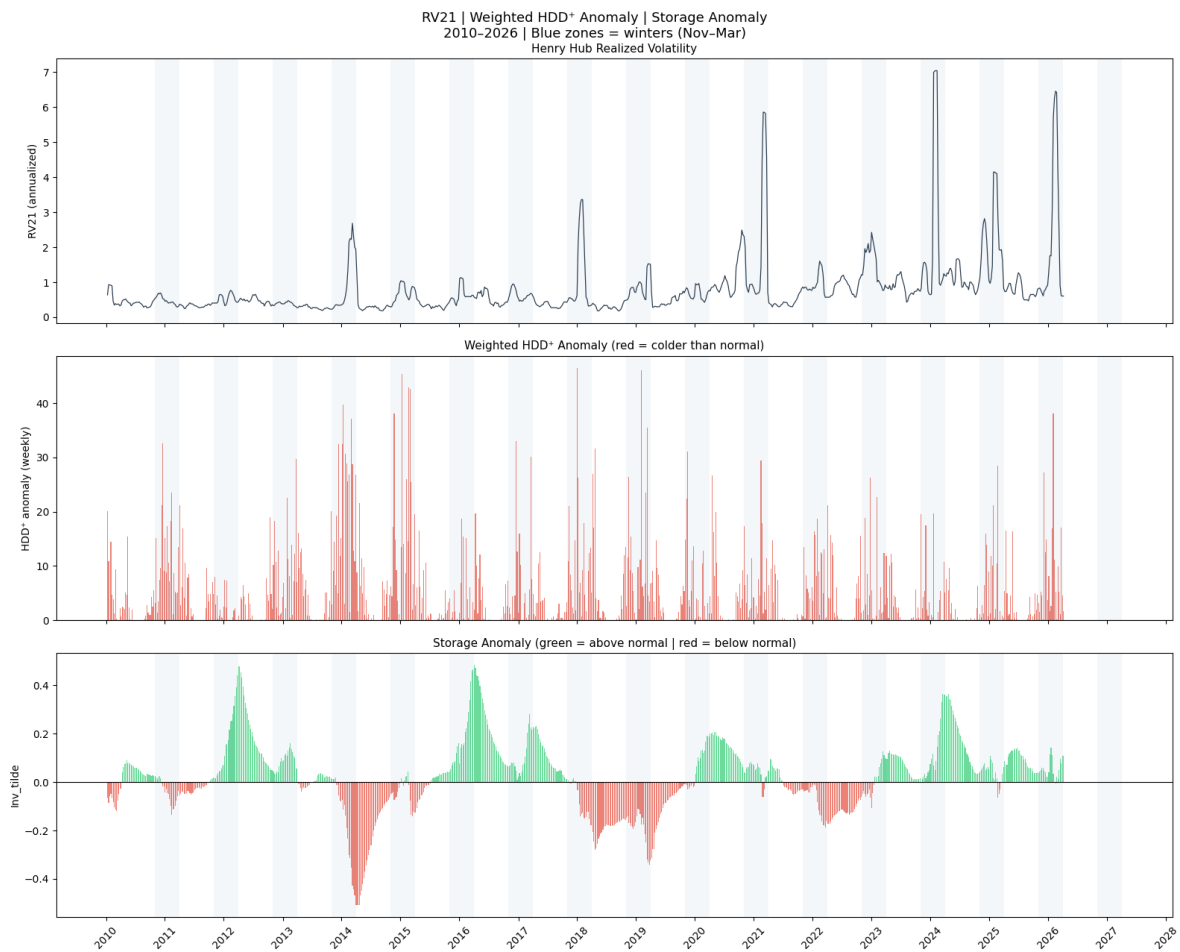


Figure 11: 2010–2026: realized volatility RV21, weighted weekly HDD⁺ anomaly, and storage anomaly $\tilde{I}_t$ relative to the weekly normal.

4.3 The Model Cascade: from Realized Anomaly to Forecast Surprise

These diagnostics guide a cascade of nested GARCH-X models, estimated by maximum likelihood, where each variant adds one hypothesis and information criteria adjudicate. Model B adds the day's composite anomaly to equation (11), $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma \widehat{\text{HDD}}_t^+$: the coefficient is significant ($\hat{\gamma} = 0.198$, $p = 0.0045$) and the AIC improves. Model C tests the storage story head-on, through an explicit interaction $\widehat{\text{HDD}}_t^+ \times \max(-\tilde{I}_t, 0)$ restricted to late winters: it fails completely ($\hat{\delta} = 0$, $p = 1.000$), partly for lack of active observations. Model D splits the anomaly between early winter (November–December) and late winter (January–March): against all expectations given figure 10, it is $\gamma_{\text{early}} = 0.618$ that stands out ($p = 0.0012$) while $\gamma_{\text{late}} = 0.190$ is not significant.

This apparent paradox is the most instructive moment of the cascade, and its resolution runs through a control variant D', an ARCH-X identical to D but *without* the memory term $\beta \sigma_{t-1}^2$: there the ranking reverses, with γ_{late} again dominant and highly significant. The explanation is that the late-winter effect is very real, but it is already *encoded in the GARCH memory*: volatility from January to March inherits, through $\beta \approx 0.86$, the cold accumulated since November, so the contemporaneous late-winter anomaly adds nothing at the margin. The coefficients of D and D' do not measure the same thing, and a naive reading of the significance levels would have led to the wrong conclusion.

The methodological consequence is immediate: to bring new information, a variable must contain what neither the memory β nor realized degree days already contain. That is the definition of a *forecast surprise*. We therefore construct, from the GEFS weather reforecasts available since 2021 for the four cities, the seven-day-ahead expectation revision $|\Delta \text{forecast}_{t,J7}|$ (figure 12). The choice of this definition is itself the outcome of a small internal contest: the signed surprise dies ($p = 1.000$, warm and cold shocks cancel out), the cold surprise works ($p = 0.0035$), but the absolute magnitude of the revision dominates ($p = 0.0001$, best AIC), a sign that it is uncertainty that moves the price, in both directions. The J+7 horizon dominates J+1, the market positioning itself on the expectation revision rather than on the morning's realization.

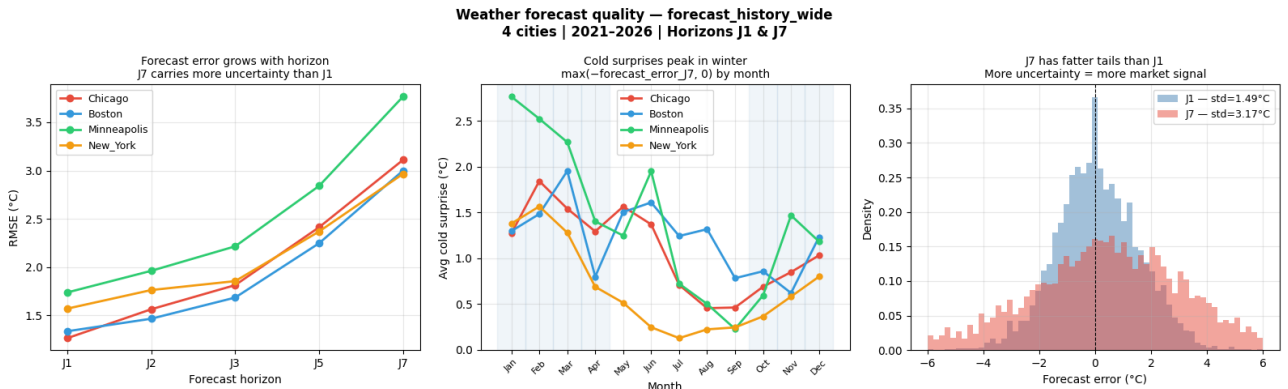


Figure 12: GEFS forecast quality for the four gas cities, 2021–2026: RMSE by horizon, average monthly cold surprise, and error distributions at J+1 and J+7.

The final model E2 combines these lessons:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma_{\text{early}} \widehat{\text{HDD}}_t^{+, \text{early}} + \gamma_{\text{late}} \widehat{\text{HDD}}_t^{+, \text{late}} + \phi |\Delta \text{forecast}_{t,J7}| \quad (12)$$

with, in sample, $\hat{\gamma}_{\text{early}} = 0.457$ ($p = 0.0004$), $\hat{\gamma}_{\text{late}} = 0.056$ (not significant, absorbed by the memory as expected), and $\hat{\phi} = 0.498$ ($p = 0.0001$).

4.4 Out-of-Sample Validation

In-sample significance is not enough: the desk's question is whether the model *forecasts* variance better. The protocol is an expanding-window backtest with annual re-estimation over 2015–2026 (2476 one-day-ahead forecasts), evaluated with the QLIKE loss, which is robust to the noise of the r_t^2 proxy, and adjudicated by the test of Diebold and Mariano (1995) with HAC variance. Table 1 gathers the results.

Table 1: Out-of-sample backtest 2015–2026 (2476 forecasts, annual refit): average QLIKE and Diebold–Mariano test against the baseline GARCH A.

Model	Overall QLIKE	Winter QLIKE	DM vs A	p (one-sided)
A: GARCH(1,1)	1.2594	1.2861	—	—
B: + $\widehat{\text{HDD}}^+$	1.2595	1.2856	−0.06	0.52
D: + early/late	1.2574	1.2807	0.78	0.22
E2: + J+7 surprise	1.2464	1.2728	3.50	0.0002

The verdict is clear-cut: realized HDD anomalies, despite their in-sample significance, add nothing in forecasting (B and D do not beat A), whereas E2 is the only model significantly better than the baseline ($DM = 3.50$, $p = 0.0002$), and remains so on the winter subsample ($DM = 2.78$) as well as on the days with a nonzero surprise alone ($DM = 3.29$). The new information channel is indeed the forecast surprise, exactly what the logic of the GARCH memory predicted.

An honest caveat closes the section. A robustness test replacing the surprise with its expanding mean (same level, no timing) or with a random permutation (timing destroyed) shows that E2 crushes the permutation but does not clearly beat the mean: part of the QLIKE gain comes from the simple seasonal uplift of the variance level. The question of what the fine *timing* of weather information is then worth, and hence a better temperature model, deserves a dedicated setup: that is the object of the next section.

5 Does a Better Temperature Model Buy Better Gas Volatility?

The main report built temperature models, both stochastic (OU) and machine learning (Ridge, Random Forest, XGBoost), and its pricing contest concluded that at the seasonal horizon, none of them delivers an economic edge over climatology. The same question now arises on the gas side, at a much shorter horizon: does the E2 cold surprise gain anything from being computed against a *better* temperature anticipation than plain climatology? A desk that pays for a good forecaster would want that skill to show up in its volatility forecast.

5.1 The design: change only the source of the anticipation

The protocol is the exact mirror of the pricing contest, transposed to volatility. We start from the full E2 model, with the GARCH dynamics, HDD controls, and backtest unchanged, and vary a single ingredient: the source of the anticipation in the one-day-ahead cold surprise,

$$\text{surprise}_{M,t} = \max(\hat{T}_{M,t} - T_t, 0), \quad M \in \{\text{clim}, \text{OU}, \text{Ridge}, \text{RF}, \text{XGB}\}, \quad (13)$$

weighted across the four gas cities by the same consumption weights. Climatology forecasts $\hat{T}_{\text{clim},t} = \theta(t)$, the OU adds the mean reversion of the residual observed the day before, $\hat{T}_{\text{OU},t} = \theta(t) + \phi_{\text{OU}} X_{t-1}$, and the machine learning models produce their one-day-ahead forecast using the true lags. Since the models are estimated on pre-2020 data, the surprises are out of sample over the entire 2020–2026 evaluation window.

5.2 Temperature skill is real, and shows up in sample

First fact, the premise of the test holds: the conditional models forecast temperature better than climatology. Over 2020–2026, their one-day-ahead RMSE ranges from 2.8 to 3.4 °C depending on the city, against 3.9 to 5.2 °C for climatology (figure 13), a gain on the order of 25 to 35 %. Their surprises are accordingly tighter and better timed (a standard deviation of 1.11 for the Random Forest against 1.67 for climatology).

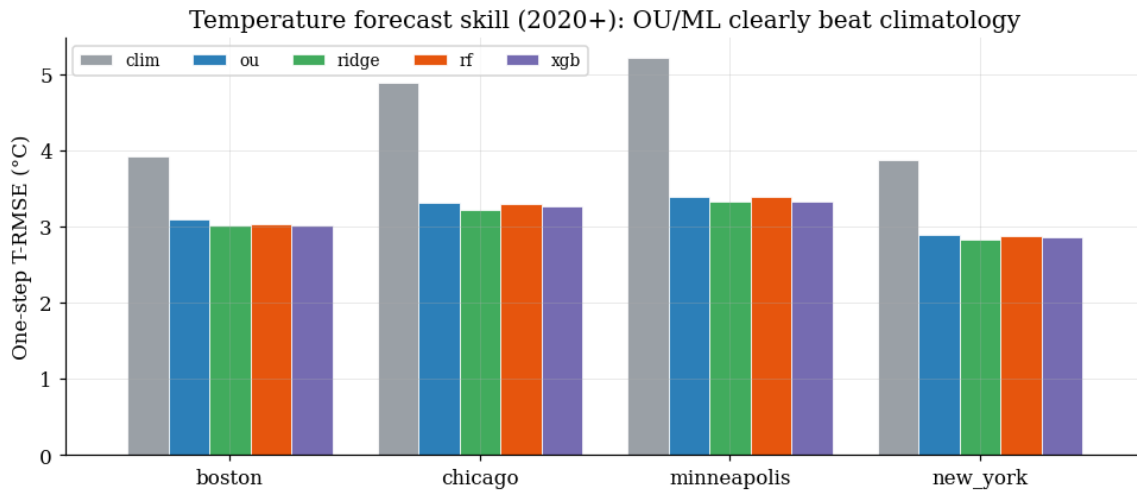


Figure 13: One-day-ahead temperature forecast RMSE (2020–2026), by city and by anticipation source.

Second fact, this skill is visible in sample: every surprise is massively significant once added to the model (LR tests between 199 and 227 against a threshold of 6.6), and the AIC ranking faithfully tracks the skill ranking, with the OU on top and climatology at the bottom. If one stopped there, the conclusion would be that a better temperature model makes a better volatility model.

5.3 Out of sample, the edge evaporates

The backtest says otherwise (figure 14). Over 2020–2026, all five surprises beat the no-surprise reference significantly (QLIKE from 1.8531 to 1.8564 against 1.8949, DM statistics between 3.0 and 3.4), but they are indistinguishable *from one another*: the maximum QLIKE gap between the best and worst source is on the order of 0.003, and the internal ranking no longer follows skill (the OU first, Ridge last, climatology in the middle). Temperature skill, though real and visible in sample, does not convert into a volatility forecasting edge.

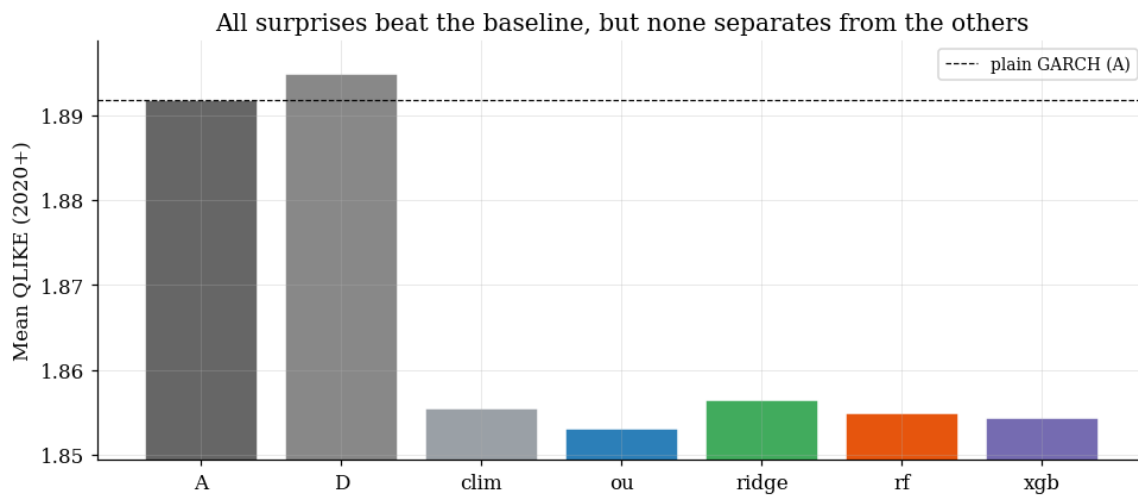


Figure 14: Average out-of-sample QLIKE (2020–2026): reference GARCH (A), HDD control (D), and the five surprise sources.

The placebos of section 4.4 explain why, by decomposing the value of the surprise into a *level* effect and a *timing* effect. For each source, the true surprise clearly beats its randomly permuted version (for the OU, $DM = 3.37$, $p = 0.0004$), so the signal is not noise; but it does not beat its expanding mean, which captures the seasonal level with no timing at all (QLIKE 1.8354 against 1.8531, $DM = -1.25$, $p = 0.89$). In other words, what the weather channel contributes to gas volatility is almost entirely the fact that winter is more volatile on average, a structure that climatology already knows perfectly. The additional precision of the OU and machine learning models, which operates day to day, has no measurable marginal value at this horizon.

This is the exact mirror of the pricing contest in the main report, and the two results reinforce each other into a unified thesis: at the horizons that respectively govern the price of a seasonal HDD contract and the conditional variance of gas, the climatological structure does the bulk of the work, and conditional temperature skill, while real, buys no economic edge either in weather option pricing or in gas volatility forecasting.

6 Conclusion

This extension has traced the footprint of temperature in the price of natural gas, in two stages. The structural route first: the two-factor Schwartz–Smith model, cast in state-space form and estimated by Kalman filter on the futures curve, would have offered the most elegant framework in which to inject weather information. It fails for a reason that is now fully documented: with maturities capped at one year, the weight of the long-term factor varies by less than 0.2% along the curve, the model is not identifiable, and the optimizer manifests this by pushing the parameters onto their bounds, up to an implied half-life of four centuries. The diagnosis is valuable precisely because it is negative: it is a data limitation, quantified and reproducible, not a limitation of the method, and a longer curve would lift it.

The volatility route then, which succeeds. A Student GARCH(1,1) captures the endogenous dynamics of the spot, and the empirical exploration pinpoints the weather effect with precision: it lives in the tails of the volatility distribution, in late winter, when storage can no longer buffer the cold. The cascade of nested models then reveals that realized degree days, although significant in sample, are already contained in the memory of the GARCH, and that the only new information channel validated out of sample is the weather forecast surprise: model E2 beats the benchmark with a Diebold–Mariano statistic of 3.50 ($p = 0.0002$), robust to winter and to surprise days.

The last word belongs to the coupling with the main report. When the OU and machine learning temperature models are plugged back into the E2 surprise, the forecasting skill, which is real and on the order of 25 to 35% lower RMSE, translates into no volatility forecasting advantage whatsoever: every source of anticipation beats the benchmark, none stands apart from the others, and the placebos show that most of the gain is a seasonal level effect that climatology already knows. This is the mirror conclusion of the HDD pricing contest, and together they form the thesis of this project: at economically relevant horizons, the climatological structure carries most of the value, and refining the temperature model, however visibly it improves the forecasts, buys no advantage on either the price of weather derivatives or the volatility of gas.

Three natural extensions emerge. Estimating the Schwartz–Smith model on a long curve (five to ten years of maturities) to lift the non-identifiability and only then connecting temperature to the short-term factor. Replacing the r_t^2 proxy with an intraday realized volatility to sharpen the assessment of timing gains. Finally, extending the framework to other hubs (the European TTF) and to power, where the weather-price link is even more direct and where the conclusion could reverse.

References

- Bollerslev, Tim (1986). “Generalized autoregressive conditional heteroskedasticity”. In: *Journal of Econometrics* 31.3, pp. 307–327. DOI: [10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1).
- Diebold, Francis X. and Roberto S. Mariano (1995). “Comparing Predictive Accuracy”. In: *Journal of Business & Economic Statistics* 13.3, pp. 253–263. DOI: [10.1080/07350015.1995.10524599](https://doi.org/10.1080/07350015.1995.10524599).
- Engle, Robert F. (1982). “Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation”. In: *Econometrica* 50.4, pp. 987–1007. DOI: [10.2307/1912773](https://doi.org/10.2307/1912773).
- Schwartz, Eduardo and James E. Smith (2000). “Short-term variations and long-term dynamics in commodity prices”. In: *Management Science* 46.7, pp. 893–911. DOI: [10.1287/mnsc.46.7.893.12034](https://doi.org/10.1287/mnsc.46.7.893.12034).

Appendices

Appendix A Moments of the Schwartz–Smith model

This appendix details the computation of the first two moments of the state vector $X_t = (\chi_t, \xi_t)^\top$ of the model in section 3.1, first in discrete time, then by passing to the continuous-time limit, and derives from them the distribution of the spot price and the futures pricing formula.

A.1 Discretization and vector form

Over a time step Δt , the Euler scheme applied to equations (2) and (3) yields

$$\chi_{t+1} = (1 - \kappa \Delta t) \chi_t + \sigma_\chi \sqrt{\Delta t} \varepsilon_\chi, \quad \xi_{t+1} = (1 - \gamma \Delta t) \xi_t + \mu_\xi \Delta t + \sigma_\xi \sqrt{\Delta t} \varepsilon_\xi,$$

with $\varepsilon_\chi, \varepsilon_\xi \sim \mathcal{N}(0, 1)$ and $\text{Cov}(\varepsilon_\chi, \varepsilon_\xi) = \rho$. Writing $a = 1 - \kappa \Delta t$ and $b = 1 - \gamma \Delta t$, the system takes the vector form

$$X_{t+1} = c + G X_t + \omega_{t+1}, \quad c = \begin{pmatrix} 0 \\ \mu_\xi \Delta t \end{pmatrix}, \quad G = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \quad W = \text{Var}(\omega) = \begin{pmatrix} \sigma_\chi^2 \Delta t & \rho \sigma_\chi \sigma_\xi \Delta t \\ \rho \sigma_\chi \sigma_\xi \Delta t & \sigma_\xi^2 \Delta t \end{pmatrix}. \quad (\text{A.1})$$

A.2 Mean and variance by recursion

Iterating the recursion equation (A.1) from X_0 ,

$$X_t = \left(\sum_{h=0}^{t-1} G^h \right) c + G^t X_0 + \sum_{i=0}^{t-1} G^i \omega_{t-i}. \quad (\text{A.2})$$

Since the noise terms are centered, the expectation retains only the first two terms. Because G is diagonal, the geometric sums can be computed component by component:

$$\mathbb{E}[X_t] = \begin{pmatrix} a^t \chi_0 \\ \mu_\xi \Delta t \frac{1-b^t}{1-b} + b^t \xi_0 \end{pmatrix}.$$

For the variance, the recursion $\text{Var}(X_t) = G \text{Var}(X_{t-1}) G^\top + W$ unrolls in the same way and, starting from $\text{Var}(X_0) = 0$,

$$\text{Var}(X_t) = \sum_{i=0}^{t-1} G^i W (G^\top)^i = \begin{pmatrix} \sigma_\chi^2 \Delta t \frac{1-a^{2t}}{1-a^2} & \rho \sigma_\chi \sigma_\xi \Delta t \frac{1-(ab)^t}{1-ab} \\ \rho \sigma_\chi \sigma_\xi \Delta t \frac{1-(ab)^t}{1-ab} & \sigma_\xi^2 \Delta t \frac{1-b^{2t}}{1-b^2} \end{pmatrix}.$$

A.3 Continuous-time limit

Fix the horizon t and set $\Delta t = t/m$ with $m \rightarrow \infty$. Then

$$a^m = \left(1 - \kappa \frac{t}{m}\right)^m \longrightarrow e^{-\kappa t}, \quad b^m \longrightarrow e^{-\gamma t}, \quad (ab)^m = \left(1 - (\kappa + \gamma) \frac{t}{m} + o\left(\frac{1}{m}\right)\right)^m \longrightarrow e^{-(\kappa + \gamma)t},$$

while to first order $1 - a^2 = 2\kappa \Delta t$, $1 - b^2 = 2\gamma \Delta t$ and $1 - ab = (\kappa + \gamma) \Delta t$. The Δt factors cancel and we obtain the continuous-time moments of equation (4):

$$\mathbb{E}[X_t] = \begin{pmatrix} e^{-\kappa t} \chi_0 \\ e^{-\gamma t} \xi_0 + \frac{\mu_\xi}{\gamma} (1 - e^{-\gamma t}) \end{pmatrix}, \quad \text{Var}(X_t) = \begin{pmatrix} \frac{\sigma_\chi^2}{2\kappa} (1 - e^{-2\kappa t}) & \frac{\rho \sigma_\chi \sigma_\xi}{\kappa + \gamma} (1 - e^{-(\kappa + \gamma)t}) \\ \cdot & \frac{\sigma_\xi^2}{2\gamma} (1 - e^{-2\gamma t}) \end{pmatrix}.$$

The log spot price, $\log S_t = \chi_t + \xi_t$, is a linear combination of a Gaussian vector: it is Gaussian, with mean and variance

$$\mathbb{E}[\log S_t] = \chi_0 e^{-\kappa t} + \xi_0 e^{-\gamma t} + \frac{\mu_\xi}{\gamma} (1 - e^{-\gamma t}), \quad \text{Var}(\log S_t) = \frac{1 - e^{-2\kappa t}}{2\kappa} \sigma_\chi^2 + \frac{1 - e^{-2\gamma t}}{2\gamma} \sigma_\xi^2 + 2 \frac{1 - e^{-(\kappa+\gamma)t}}{\kappa + \gamma} \sigma_\chi \sigma_\xi \rho.$$

A.4 Log-normality and futures prices

The spot price $S_t = e^{\log S_t}$ is log-normal. For $X = \log S_t \sim \mathcal{N}(\mu, \sigma^2)$,

$$\mathbb{E}[S_t] = \mathbb{E}[e^X] = \int_{\mathbb{R}} e^x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = e^{\mu + \frac{\sigma^2}{2}} \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du = e^{\mu + \frac{\sigma^2}{2}},$$

where we completed the square in the numerator and then substituted $u = (x - (\mu + \sigma^2))/\sigma$, the remaining Gaussian integral being equal to one.

Under the risk-neutral measure $\mathbb{Q}$, the same calculations apply to the risk-premium-adjusted dynamics, $d\chi_t = (-\kappa\chi_t - \lambda_\chi)dt + \sigma_\chi dZ_t^{\chi, \mathbb{Q}}$ and $d\xi_t = (\mu_\xi - \lambda_\xi - \gamma\xi_t)dt + \sigma_\xi dZ_t^{\xi, \mathbb{Q}}$. Since the price of a futures contract with maturity T is by definition $F(t, T) = \mathbb{E}_t^{\mathbb{Q}}[S_T]$, log-normality gives

$$\log F(t, T) = \mathbb{E}_t^{\mathbb{Q}}[\log S_T] + \frac{1}{2} \text{Var}_t^{\mathbb{Q}}(\log S_T) = e^{-\kappa(T-t)}\chi_t + e^{-\gamma(T-t)}\xi_t + A(T-t),$$

with $A(\tau)$ given by equation (7), which establishes equation (6).

A.5 Exact discretization for the filter

For estimation, we prefer to the Euler scheme the *exact* discretization, obtained by solving the SDEs factor by factor (the same method as in the temperature pillar):

$$\chi_t = e^{-\kappa\Delta t} \chi_{t-1} + \sigma_\chi \int_{t-1}^t e^{-\kappa(t-s)} dZ_s^\chi, \quad \xi_t = \frac{\mu_\xi}{\gamma} (1 - e^{-\gamma\Delta t}) + e^{-\gamma\Delta t} \xi_{t-1} + \sigma_\xi \int_{t-1}^t e^{-\gamma(t-s)} dZ_s^\xi.$$

We recover the state-space form equation (8) with $G = \text{diag}(e^{-\kappa\Delta t}, e^{-\gamma\Delta t})$, and the state noise ω_t has exact covariance

$$W = \begin{pmatrix} \frac{\sigma_\chi^2}{2\kappa} (1 - e^{-2\kappa\Delta t}) & \frac{\rho \sigma_\chi \sigma_\xi}{\kappa + \gamma} (1 - e^{-(\kappa+\gamma)\Delta t}) \\ \cdot & \frac{\sigma_\xi^2}{2\gamma} (1 - e^{-2\gamma\Delta t}) \end{pmatrix},$$

the cross term following from Itô's isometry applied to the product of the two stochastic integrals, $\text{Cov}(\chi_t, \xi_t | \mathcal{F}_{t-1}) = \rho \sigma_\chi \sigma_\xi \int_{t-1}^t e^{-\kappa(t-s)} e^{-\gamma(t-s)} ds$.

Appendix B Derivation of the Kalman filter

This appendix derives the Kalman filter equations used in section 3.3, for the state-space system

$$X_t = c + G X_{t-1} + \omega_t, \quad y_t = d_t + F_t^\top X_t + v_t, \quad (\text{B.1})$$

where $X_t \in \mathbb{R}^2$ is the latent (unobserved) state, $y_t \in \mathbb{R}^M$ collects the log futures prices, $\omega_t \sim \mathcal{N}(0, W)$ is the state noise and $v_t \sim \mathcal{N}(0, V)$ the measurement noise. The working assumptions are as follows: linearity of the system (both equations are affine in X_t), Gaussian noises, independence of the noises from each other and over time ($\omega_t \perp v_s$ for all t, s , $\omega_t \perp \omega_s$ and $v_t \perp v_s$ for $t \neq s$), known covariances W and V (in practice

parameterized by Θ and estimated by maximum likelihood, the filter being applied conditionally on Θ at each iteration of the optimizer), and a specified initialization a_0, P_0 . Without Gaussianity, the filter remains the best *linear* estimator (generalized Gauss–Markov theorem), but Gaussianity guarantees global optimality.

Let $\mathcal{F}_t$ denote the information generated by the observations up to date t , and define the filter quantities

$$a_{t|t-1} = \mathbb{E}[X_t | \mathcal{F}_{t-1}], \quad a_t = \mathbb{E}[X_t | \mathcal{F}_t], \quad P_{t|t-1} = \text{Cov}(X_t | \mathcal{F}_{t-1}), \quad P_t = \text{Cov}(X_t | \mathcal{F}_t), \quad L_{t|t-1} = \text{Cov}(e_t | \mathcal{F}_{t-1}),$$

where $e_t = y_t - \mathbb{E}[y_t | \mathcal{F}_{t-1}]$ is the one-step prediction error on the observation. By induction, X_t is a linear combination of independent Gaussians, so y_t and $y_t | \mathcal{F}_{t-1}$ are Gaussian: all the conditional distributions of the filter are Gaussian and fully described by these moments.

B.1 Prediction

Applying the conditional expectation and covariance to the state equation, with $\omega_t \perp \mathcal{F}_{t-1}$:

$$a_{t|t-1} = c + G a_{t-1}, \quad P_{t|t-1} = \text{Cov}(GX_{t-1} + \omega_t | \mathcal{F}_{t-1}) = G P_{t-1} G^\top + W. \quad (\text{B.2})$$

B.2 Innovation

The prediction error on the observation is obtained by subtracting $\mathbb{E}[y_t | \mathcal{F}_{t-1}] = d_t + F_t^\top a_{t|t-1}$ (the measurement noise being centered and independent of $\mathcal{F}_{t-1}$):

$$e_t = y_t - d_t - F_t^\top a_{t|t-1} = F_t^\top (X_t - a_{t|t-1}) + v_t. \quad (\text{B.3})$$

Its conditional covariance expands into four terms. The two cross terms vanish because v_t is independent of X_t and of $\mathcal{F}_{t-1}$, so that $\mathbb{E}[(X_t - a_{t|t-1}) v_t^\top | \mathcal{F}_{t-1}] = \mathbb{E}[X_t - a_{t|t-1} | \mathcal{F}_{t-1}] \mathbb{E}[v_t]^\top = 0$. This leaves the state term and the measurement term:

$$L_{t|t-1} = F_t^\top \mathbb{E}[(X_t - a_{t|t-1})(X_t - a_{t|t-1})^\top | \mathcal{F}_{t-1}] F_t + \mathbb{E}[v_t v_t^\top] = F_t^\top P_{t|t-1} F_t + V. \quad (\text{B.4})$$

Hence $e_t | \mathcal{F}_{t-1} \sim \mathcal{N}(0, L_{t|t-1})$.

B.3 Correction and Kalman gain

Once y_t is observed, the prediction is corrected linearly in the error: $a_t = a_{t|t-1} + K_t e_t$, where the gain matrix K_t is chosen to minimize the mean squared error $\mathbb{E}[\|X_t - a_t\|^2 | \mathcal{F}_t] = \text{tr}(P_t)$. Setting $\Sigma_t = X_t - a_{t|t-1}$, the post-correction error reads

$$X_t - a_t = \Sigma_t - K_t e_t = (I - K_t F_t^\top) \Sigma_t - K_t v_t,$$

and expanding the product $(X_t - a_t)(X_t - a_t)^\top$, the cross terms in $\Sigma_t v_t^\top$ vanish in expectation (same independence arguments as above). What remains is

$$P_t = (I - K_t F_t^\top) P_{t|t-1} (I - K_t F_t^\top)^\top + K_t V K_t^\top = P_{t|t-1} - K_t F_t^\top P_{t|t-1} - P_{t|t-1} F_t K_t^\top + K_t L_{t|t-1} K_t^\top, \quad (\text{B.5})$$

where we have grouped $K_t (F_t^\top P_{t|t-1} F_t + V) K_t^\top = K_t L_{t|t-1} K_t^\top$. The first-order condition, obtained using the matrix differentiation rules $\partial_K \text{tr}(KA) = A^\top$ and $\partial_K \text{tr}(KAK^\top) = 2KA$ for symmetric A , yields

$$\frac{\partial \text{tr}(P_t)}{\partial K_t} = -2 P_{t|t-1} F_t + 2 K_t L_{t|t-1} = 0 \iff \boxed{K_t = P_{t|t-1} F_t L_{t|t-1}^{-1}} \quad (\text{B.6})$$

Substituting equation (B.6) back into equation (B.5), the last two terms partially cancel and the updated covariance takes its usual form:

$$\boxed{a_t = a_{t|t-1} + K_t e_t, \quad P_t = (I - K_t F_t^\top) P_{t|t-1}} \quad (\text{B.7})$$

The gain K_t calibrates the strength of the correction: if the prediction uncertainty $P_{t|t-1}$ dominates the measurement noise V , the gain is large and the observation prevails, and conversely.

B.4 Likelihood via the prediction error decomposition

The joint distribution of the observations factorizes into a product of conditionals, $\mathcal{L}(\Theta; y) = \prod_{t=1}^{M_T} p(y_t | \mathcal{F}_{t-1})$, and each conditional is Gaussian, $y_t | \mathcal{F}_{t-1} \sim \mathcal{N}(d_t + F_t^\top a_{t|t-1}, L_{t|t-1})$. The log-likelihood therefore writes directly in terms of the filter outputs:

$$\ell(\Theta; y) = -\frac{1}{2} \sum_{t=1}^{M_T} \left[M \log(2\pi) + \log \det(L_{t|t-1}) + e_t^\top L_{t|t-1}^{-1} e_t \right], \quad (\text{B.8})$$

each pass of the filter at date t contributing its own term. Estimation consists in numerically maximizing ℓ over Θ , the filter being rerun at each evaluation.

Appendix C Non-identifiability with a short futures curve

This appendix formalizes the argument of section 3.4. Recall the definition: a model parameterized by Θ is *identifiable* if $p(y; \Theta_1) = p(y; \Theta_2)$ implies $\Theta_1 = \Theta_2$. If two distinct parameter sets produce exactly the same distribution of the observations, no amount of data can tell them apart and the maximum likelihood estimator is not consistent: it does not converge to any point, the likelihood exhibiting a flat direction.

C.1 Degeneracy of the model as $\gamma \rightarrow 0$

Start from the measurement equation equation (9). The observation matrix stacks, for each maturity τ_i , the weights of the two factors:

$$F_t = \begin{pmatrix} e^{-\kappa \tau_1} & \dots & e^{-\kappa \tau_M} \\ e^{-\gamma \tau_1} & \dots & e^{-\gamma \tau_M} \end{pmatrix} \xrightarrow{\gamma \rightarrow 0} \begin{pmatrix} e^{-\kappa \tau_1} & \dots & e^{-\kappa \tau_M} \\ 1 & \dots & 1 \end{pmatrix}:$$

the row of the long-term factor becomes constant, with all maturities loading on it identically. In the same regime, the component of $A(\tau)$ carried by the drift becomes linear in maturity, $\frac{\mu_\xi - \lambda_\xi}{\gamma} (1 - e^{-\gamma \tau}) \rightarrow (\mu_\xi - \lambda_\xi) \tau$, and each observation can be written as

$$y_{t,i} = \left(\mu_\xi - \lambda_\xi + \frac{1}{2} \sigma_\xi^2 \right) \tau_i + g(\tau_i) + e^{-\kappa \tau_i} \chi_t + \xi_t + v_{t,i}, \quad (\text{C.1})$$

where $g(\tau_i)$ collects the terms in κ , σ_χ , σ_ξ , ρ and λ_χ , which depend neither on the drift nor on the level of the long-term factor.

Consider first a curve reduced to a single maturity τ . For any $\delta \neq 0$, the translated parameter set

$$\Theta_2: \quad \mu_\xi \mapsto \mu_\xi + \delta, \quad \xi_t \mapsto \xi_t - \delta \tau,$$

leaves equation (C.1) unchanged, term by term: the additional slope $\delta \tau$ is exactly absorbed by the drop in the level. Since the noise $v_{t,i}$ is the same in both formulations, $p(y; \Theta_1) = p(y; \Theta_2)$ for every trajectory: there exists

a continuous family of pairs (μ_ξ, ξ_t) producing exactly the same observations. The model is not identifiable and the maximum likelihood estimator does not converge in probability to an optimal parameter.

With several maturities, the translation above depends on τ_i and no longer defines a common state: the exact identity breaks down, and it is precisely the spread of maturities that must identify the factor. But if the observed range $[\tau_1, \tau_M]$ is narrow relative to $1/\gamma$, the weights $e^{-\gamma\tau_i}$ are nearly constant across maturities and the direction $(\mu_\xi + \delta, \xi_t - \delta\bar{\tau})$ changes the likelihood only at second order, by an amount proportional to the dispersion of the τ_i around their mean $\bar{\tau}$. The likelihood then exhibits a nearly flat valley: theoretical identifiability in the strict sense, but inference that is impossible in practice, with the estimates drifting freely along the valley at the whim of the measurement noise.

C.2 Orders of magnitude: the condition on maturities

Identifying the two factors requires that their weights vary detectably along the observed curve, which imposes two conditions on the maximum maturity $\tau_{\max}$:

$$(i) \quad e^{-\gamma\tau_{\max}} \ll 1 \quad \Longleftrightarrow \quad \tau_{\max} \gg \frac{1}{\gamma}, \quad (ii) \quad e^{-\kappa\tau_{\max}} \approx 0 \quad \Longleftrightarrow \quad \tau_{\max} \gg \frac{1}{\kappa}.$$

Condition (ii) ensures that long contracts are informative only about ξ_t ; condition (i) ensures that the decay of the weight of ξ_t is visible, making it possible to separate level and drift. Short contracts, by difference with long contracts, are then informative about χ_t .

With the orders of magnitude of natural gas, $\gamma \approx 0.1$ (a long-term factor half-life of several years) and $\kappa \approx 1.3$ (shocks absorbed within a few months), and our Databento chain with maximum maturity $\tau_{\max} = 1$ year:

$$e^{-\gamma\tau_{\max}} = e^{-0.1} = 0.905, \quad e^{-\kappa\tau_{\max}} = e^{-1.3} = 0.272.$$

For all our maturities, $e^{-\gamma\tau_i} \in [0.905, 1]$: the total variation of the long-term factor's weight along the curve is below ten percent, indistinguishable from the measurement noise. Condition (ii) is approximately satisfied ($1/\kappa \approx 0.8$ year), while condition (i) would require $\tau_{\max} \gg 1/\gamma \approx 10$ years. This is a data problem, not a methodological one: Schwartz and Smith (2000) calibrate their model on futures with maturities of up to nine years, whereas our chain stops at one year.

Figure C.1 shows the optimizer trace for the best starting point: the log-likelihood stabilizes while κ (denoted zeta in the implementation) and γ drift toward their respective lower bounds, without ever finding an interior minimum, the numerical manifestation of the flat valley described above.

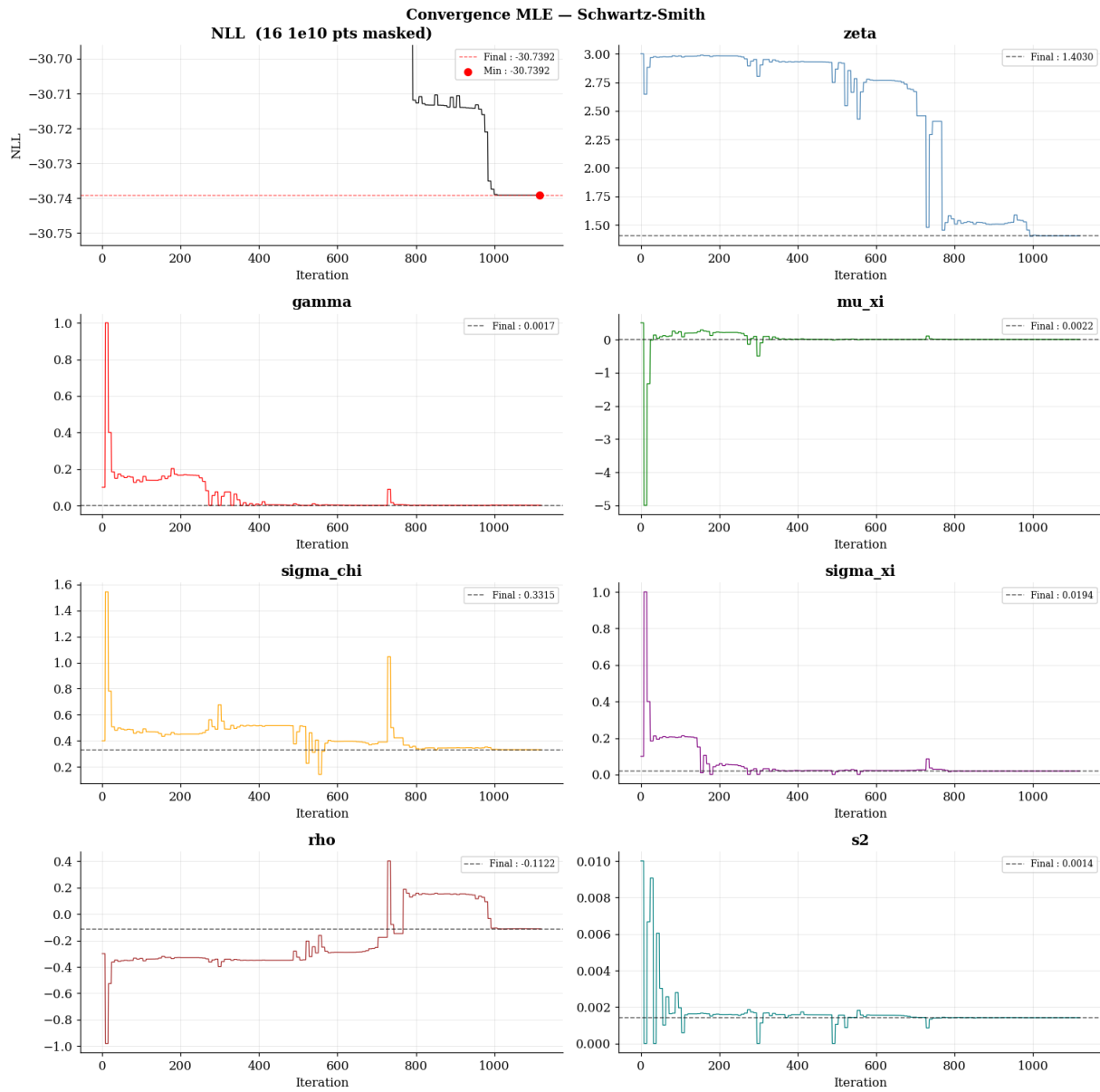


Figure C.1: Trace of the MLE optimization of the Schwartz–Smith model (best initialization): log-likelihood and parameter trajectories across iterations.